

Corporate Liquidity and Debt Buybacks

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Discussion by:

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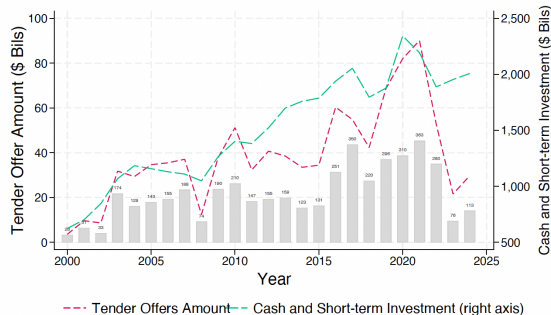
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Context: Debt Buybacks

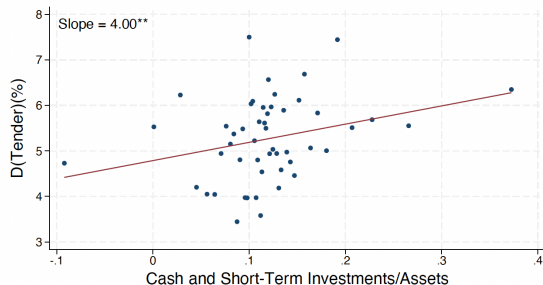
Firms retire their bonds before maturity, and they pay the market price.

- 4,347 verified tender offers, \$1.08 trillion retired over 2000–2024

(a) Aggregate Tender Offer Activity



(b) Tender Offer Likelihood



In the data, corporate bond buybacks track corporate cash.

Paper in One Slide

RQ: Why do firms buy back their corporate debt?

The literature suggests that they shouldn't (Admati et al. 2018; DeMarzo and He 2021).

- Leverage ratchet effect: equity holders have an incentive to increase debt but are reluctant to reduce it.

DeMarzo and He (2021) (DH) show that equity holders issue debt continuously and **never buy it back**, no matter how levered the firm is.

This paper **generates buybacks** in a DH-type model by adding **cash liquidity concern**.

- **key friction**: firms cannot freely borrow → liquidation is costly

Empirically, data confirm the model's predictions: cash-rich firms generally buy back more, and firms adopt debt and dividend payout policies as the model predicts.

Static Model

- Two period ($t = 0, 1$), no discounting
- The firm needs to repay debt of F from cash flows $\tilde{W}$ at $t = 1$.
 - if $W \geq F$, the firm repays debt from cash;
 - if $W < F$, it can either continue by liquidating assets or default.

1. Liquidating partially will damage the going-concern value

$$A + \underbrace{\theta(W - F)}_{<0}, \quad \theta > 1.$$

2. Default gives the cash back to equity holders:

$$W$$

- Continue iff

$$\theta(W - F) + A \geq W \quad \Rightarrow \quad \boxed{W \geq W_d \equiv \frac{\theta F - A}{\theta - 1}.$$

Static Model: Equity Payoff

- Date-1 equity payoff:

$$V^1(W; F) = \begin{cases} W, & W < W_d, \\ \theta(W - F) + A, & W_d \leq W < F, \\ W - F + A, & W \geq F. \end{cases} \quad (2)$$

- Marginal value of cash:

$$V_W^1(W; F) = \begin{cases} 1, & W < W_d, \\ \theta, & W_d \leq W < F, \\ 1, & W \geq F. \end{cases}$$

- Cash has a liquidity premium (for **shareholders**) only in the **partial-liquidation** region:

$$V_W = \theta > 1.$$

Static Model: Debt Pricing and Date-0 Equity Value

- Define $d \equiv \Pr(\tilde{W} < W_d)$, $\ell \equiv \Pr(W_d \leq \tilde{W} < F)$, $s \equiv \Pr(\tilde{W} \geq F)$.
- Risk-neutral lenders price debt while considering the default risk

$$p(F) \equiv \Pr(\text{do not default}) = \Pr(\tilde{W} \geq W_d) = \ell + s < 1.$$

- Date-0 equity value:

$$V^0(F) = \mathbb{E}[V^1(\tilde{W}; F)].$$

- The marginal value of reducing debt is:

$$\boxed{-V_F^0 = \theta \ell + s.} \tag{5}$$

Intuition:

retiring one unit of face is worth

0 in default states, θ in partial-liquidation states, 1 in survival states.

Static Model: Intuition

Q: What is the marginal value of hoarding an additional dollar ($W \rightarrow W + \Delta w$) on the B/S?

$$\left. \frac{\partial V^{0,\text{hoard}}}{\partial w} \right|_{w=0} = d \cdot 1 + \ell \cdot \theta + s \cdot 1 = 1 + (\theta - 1)\ell \geq 1. \quad (7)$$

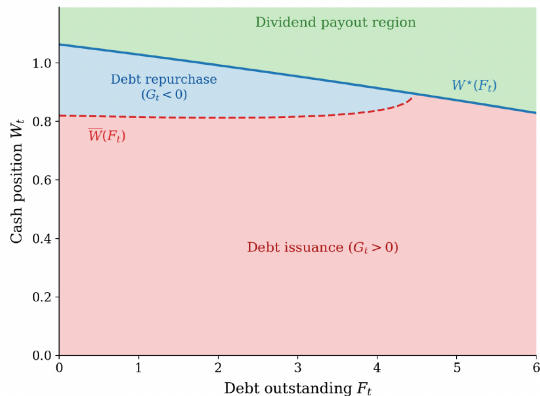
Q: What is the marginal value of spending a dollar on buybacks?

$$\text{one dollar retires } \frac{1}{p} \text{ units of debt} \times (-V_F^0) = \frac{\theta \ell + s}{\ell + s}$$

⇒ **Buyback is more valuable** when $\theta > 1$.

$$\frac{\theta \ell + s}{\ell + s} - [1 + (\theta - 1)\ell] = \frac{d}{1 - d} \ell (\theta - 1) > 0.$$

Dynamic Model: Optimal Debt Policy



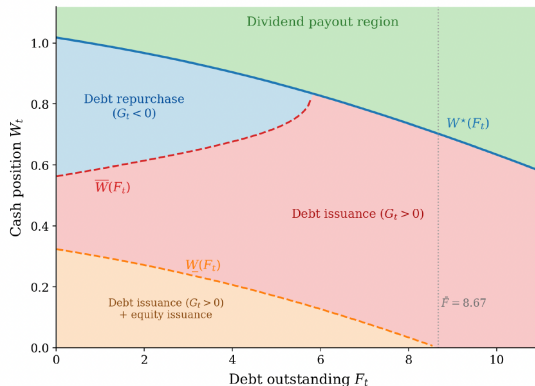
Below the dividend payout boundary, $V_W > 1$: never pay out dividends

- Cash buffer provides valuable insurance against liquidity-driven bankruptcy.

Cash is relatively abundant: buy back debt, since the cash-carry wedge dominates.

Dynamic Model that Allows for Equity

Main message: debt buyback mechanism **still holds** with external equity financing.



When do firms issue equity? **when cash is scarce and when debt is not too high**

Empirical Evidence

- Sample: all US corporate bonds from FISD from 2000 to 2024
 - firm-year panel: 31,705 firm-years, 45, 109 bonds
- TCJA as a **liquidity shock**: it released foreign-held cash for domestic use.
 - before TCJA: U.S. multinationals had tax incentives to keep foreign cash abroad;
 - TCAJ: made repatriation much cheaper at the margin
- The dynamic model predicts a state-space payout:
 - $\bar{W}(F) < W < W^*(F) \Rightarrow$ debt buyback
 - $W \geq W^*(F) \Rightarrow$ equity payout
- Sort firms into quartiles by cash shock (average foreign income-to-sales):
 - Q3-windfall moves firms into buyback region
 - Q4-windfall pushes firms past payout boundary

Empirical Evidence

	1(Tender Offers)(%)		Frac(Tender Offers)(%)		Frac(Equity Repurchase)(%)	
	(1)	(2)	(3)	(4)	(5)	(6)
Quartile 3 PIFO $\times$ Post	4.714** (1.990)	4.799** (1.990)	0.699** (0.324)	0.709** (0.324)	0.071 (0.303)	0.047 (0.301)
Quartile 4 PIFO $\times$ Post	3.158* (1.841)	3.087* (1.841)	0.279 (0.286)	0.272 (0.286)	0.543** (0.275)	0.574** (0.277)
Leverage(t-1)		6.105** (2.908)		0.721 (0.527)		-2.301*** (0.477)
Adjusted R^2	0.103	0.103	0.054	0.054	0.288	0.295
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
NAICS3-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,598	5,571	5,595	5,568	5,468	5,453

Moderate liquidity shocks $\Rightarrow$ bond buybacks; Large liquidity shocks $\Rightarrow$ equity payouts.

Outline

Comment 1: Alternative Explanations

Comment 2: What about nonsmooth equilibria?

Comment 1: Motivation and Framing

- This paper is currently framed around the **non-buyback puzzle** generated in DeMarzo and He (2021).
- Indeed, in a very skeptical world with no commitment, adding liquidity friction can generate buybacks.
- However, there are already alternative mechanisms to generate buybacks.
- For example, Malenko and Tsoy (2026) show the reputation concern can restore commitment and generate buybacks.
 - Markov perfect equilibrium vs. non-MPE
- **Suggestion:** downplay the puzzle and emphasize the importance of **a general framework featuring both cash liquidity and debt** without commitment.

Outline

Comment 1: Alternative Explanations

Comment 2: What about nonsmooth equilibria?

Optimality Condition and Value Function

Solving for a smooth equilibrium is already difficult.

For a given state (F, W) , issuing a small amount dF produces two effects:

$$\underbrace{V_F dF}_{\text{effect of the additional debt}} + \underbrace{V_W \cdot p dF}_{\text{value of the cash raised}} .$$

For the firm to adjust debt at a finite rate in a smooth equilibrium, these must balance:

$$V_F + pV_W = 0.$$

Idea: given the equilibrium price and continuation value, shareholders have no first-order gain from deviating from current debt right now.

Optimality Condition and Value Function

Value function for the equity holders:

$$V(F_t, W_t) = \max_{\{\Gamma_s, D_s\}_{s \geq t}} \mathbb{E}_t \left[\int_t^{\tau \wedge T} e^{-\rho(s-t)} dD_s + e^{-\rho(T-t)} \mathbf{1}_{\{T < \tau\}} \widehat{V}(F_T, W_T) \right]. \quad (12)$$

Rewrite $d\Gamma_s \equiv G_t dt$, under the region where the firm doesn't pay out dividends, the HJB is

$$\rho V(F, W) = \max_G \left\{ G \mathbf{V}_F + [(1 - \pi)(\mu - cF) + pG] \mathbf{V}_W + \frac{1}{2}(1 - \pi)^2 \sigma^2 V_{WW} + \lambda [\widehat{V}(F, W) - V(F, W)] \right\}. \quad (23)$$

We can characterize the value function, without even knowing debt issuance and buyback:

$$(\rho + \lambda)V = (1 - \pi)(\mu - cF)V_W + \frac{1}{2}(1 - \pi)^2 \sigma^2 V_{WW} + \lambda W. \quad (24)$$

- For a given F , this is an ODE for V w.r.t. W !
- ⇒ solve for the equity holders' value function $V(F, W)$ ✓
- this technique has also been used in DeMarzo and He (2021)

Optimal Debt Policy

Having characterized the value function $V(F, W)$, the optimal debt policy is obtained from the equilibrium condition:

$$G_t = \frac{\overbrace{\pi C}^{\text{tax benefit}} - \overbrace{\left[\rho + \lambda \frac{V_W - 1}{V_W} \right] p}^{\text{cash-carry wedge}} + \overbrace{(1 - \pi)^2 \sigma^2 \cdot p_W \cdot \frac{-V_{WW}}{V_W}}^{\text{precautionary liquidity}}}{-(p_F + p \cdot p_W)} = \frac{\text{net benefits}}{\text{adverse price impact}}. \quad (34)$$

1. Tax benefit: coupon payments are tax deductible $\rightarrow$ debt tax shield benefit flow
2. Cash-carry wedge **[Why equity holders might want to repurchase]**
 - ρ : equity holders are impatient, issuing an extra unit of debt without distributing cash (on the B/S) is a cost
 - $\frac{\lambda(V_W - 1)}{V_W}$: termination reduces the cash value from V_W to 1 with prob. $\lambda \rightarrow$ termination destroys the liquidity premium
3. Precautionary motive: equity value is concave in cash ($V_{WW} < 0$) because cash is more valuable when the firm is closer to running out of cash

Comment 2: Solution Logic and Nonsmooth Eq

This paper uses a **conjecture-and-verify** method to construct a smooth equilibrium.

- i. suppose the firm **only adjusts debt smoothly**, we can construct a smooth equilibrium
- ii. in the smooth eq, indeed **smooth debt adjustment is optimal** (*Lemma 2*)

What if the firm can adjust the debt rapidly? The paper doesn't offer a clear overall picture.

Section 4.2 shows both the **robustness** and **limitation** of smooth eq:

- Robustness: if outside equity is too expensive, the baseline smooth eq remains valid
- Limitation: if equity issuance can be nonsmooth, the smooth debt adjustment eq **cannot sustain**

⇒ Q: What would nonsmooth equilibrium look like with cash management?

Recap

- A very solid theoretical paper with supporting empirical evidence
- **Punchline:** Debt buybacks are **puzzling** under leverage-ratchet logic $\Rightarrow$ **incorporating the liquidity friction** resolves this puzzle.
- A promising research agenda: future research could further explore the interaction between **cash** and other frictions, such as **transaction costs, reputation concerns, or agency problems**.
- I learned a lot from reading this paper!